## Halward Inspection Services Ltd.

**Inspection Report**

Site: Hamilton Plant

| No. | Status | Date |
|---|---|---|
| HIS-2007-0388 | Issued | 2007-09-18 |

# UT Thickness Survey and Fitting Scans - Regenerator Overhead Circuit

## Scope and Responsibilities

This report details the ultrasonic thickness survey conducted on the regenerator overhead circuit within the ATU-100. The scope includes a comprehensive assessment of every CML on lines 3-RV-101 and 2-RV-102, supplemented by high-density grid scans of all associated fittings. Kevin Doucette served as the lead UT technician for this campaign, ensuring all data acquisition met the requirements for SNT-TC-1A UT Level II certification.

Halward Inspection Services Ltd. executed the field measurements in accordance with HIS-UT-002 Rev 1. All activities were further governed by the site-specific requirements of QSCP-NDT-001 Rev 0. The inspection was prioritized following the issuance of QSCP-WP-2007-09311, which documented a pinhole leak at a 90° elbow on 2-RV-102 where no previous CML existed, necessitating a full circuit evaluation.

## Inspection Methodology

Thickness data was collected using digital ultrasonic equipment following the contact pulse-echo method described in ASTM E797. Surface preparation was performed at each CML to ensure reliable coupling. For the fittings, a grid pattern was established to identify localized thinning, particularly at the extrados of elbows where flow-accelerated corrosion or erosion-corrosion is suspected.

The survey adhered to the inspection practices outlined in API 570 for piping systems. Measurements were compared against the original nominal thicknesses of 7.62 mm for 3-RV-101 and 5.54 mm for 2-RV-102. All calibration was verified against step blocks of similar A106 Gr.B material to maintain accuracy across the design temperature range of 121 °C.

## Data Analysis and Site Rules

Evaluation of the results follows the QSCP Hamilton Plant site rules for asset integrity. The retirement thickness is defined as the t-min for each line, specifically 3.56 mm for 3-RV-101 and 2.79 mm for 2-RV-102. Any reading found below these values is immediately raised as an NCR. Corrosion rates are determined by selecting the larger of the long-term rate and the short-term rate, provided the latest readings are at least 3 years apart.

Remaining life is calculated by dividing the available corrosion allowance by the determined rate. As these lines are part of the regenerator overhead circuit carrying sour gas, they are designated as Class 1. Consequently, the next thickness reading must be scheduled at no more than half the remaining life, not to exceed the 5-year class maximum for thickness and external visual inspections.

## Summary of Findings

The grid scans performed under QSCP-WP-2007-09311 confirmed that the localized thinning at the 2-RV-102 elbow was not isolated to the leak site. While the majority of the circuit remains above t-min, the accelerated corrosion observed in the 2-inch section requires a reassessment of the current CML placement. The 3-RV-101 line showed more uniform wall loss consistent with design expectations for 345 kPa(g) service.

All data has been recorded in the plant asset management system to update the short-term corrosion rate calculations. The findings from this survey by Kevin Doucette will be used to adjust the inspection intervals for the ATU-100 overhead circuit to ensure continued compliance with API 570 and site safety standards.

## Thickness readings, measured 2007-09-18

**Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RV-101-01 | 7.62 | 3.56 | 2006-02-24 | 6.98 | 6.95 | 0.056 | 0.047 | >50 | 2012-09-18 | |
| 3-RV-101-02 | 7.62 | 3.56 | 2006-02-24 | 6.93 | 6.89 | 0.059 | 0.064 | >50 | 2012-09-18 | |
| 3-RV-101-03 | 7.62 | 3.56 | 2006-02-24 | 6.57 | 6.56 | 0.083 | 0.090 | 33.3 | 2012-09-18 | |
| 3-RV-101-04 | 7.62 | 3.56 | 2006-02-24 | 6.65 | 6.66 | 0.081 | 0.079 | 38.4 | 2012-09-18 | |
| 3-RV-101-05 | 7.62 | 3.56 | 2006-02-24 | 7.01 | 6.87 | 0.069 | 0.058 | 47.7 | 2012-09-18 | |
| 3-RV-101-06 | 7.62 | 3.56 | 2006-02-24 | 6.74 | 6.71 | 0.076 | 0.075 | 41.3 | 2012-09-18 | |

**Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RV-102-01 | 5.54 | 2.79 | 2006-02-24 | 4.85 | 4.83 | 0.061 | 0.070 | 29.0 | 2012-09-18 | |
| 2-RV-102-02 | 5.54 | 2.79 | 2006-02-24 | 4.84 | 4.67 | 0.080 | 0.069 | 23.6 | 2012-09-18 | |
| 2-RV-102-03 | 5.54 | 2.79 | 2006-02-24 | 5.14 | 5.01 | 0.041 | 0.035 | >50 | 2012-09-18 | |
| 2-RV-102-04 | 5.54 | 2.79 | 2006-02-24 | 5.07 | 5.12 | 0.040 | 0.032 | >50 | 2012-09-18 | |

| Sign-off | Name | Function | Date | Signature |
|---|---|---|---|---|
| Prepared | Kevin Doucette | lead UT technician, SNT-TC-1A UT Level II | 2007-09-18 | _____ |